Bell Cyber

NIST Mapping (SP 800-53 Rev. 5 and CSF 2.0)

Companion mapping to the Bell Cyber Secure Code Review of Splunk security_content

Field	Value
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NIST Mapping (SP 800-53 Rev. 5 and CSF 2.0)

This document maps every finding from the Bell Cyber secure code review of the Splunk `security_content` repository to the NIST SP 800-53 Revision 5 controls and the NIST Cybersecurity Framework 2.0. It is a companion to the full report; the finding identifiers (for example PB2-01) resolve to the detailed write-ups in Section 4 of that report. The finding catalogue is repeated below for reference, followed by the mapping.

Finding catalogue

ID	Title	Severity	CVSS	Area
DEP-02	Unpinned pip install in a scheduled job holding long-lived AWS keys that writes the shared app-binaries bucket	High	8.6	Dependencies
PB1-02	CrowdStrike RTR command-argument injection and path traversal via raw path/pid inputs	High	8.5	SOAR playbooks
PB2-03	PowerShell command injection via unquoted filepath in internal_host_winrm_log4j_respond	High	8.3	SOAR playbooks
PB1-01	CrowdStrike FQL filter injection via playbook_input:device (10 playbooks)	High	8.2	SOAR playbooks
CI-01	Script injection via github.head_ref interpolated into run: steps (unit-testing workflow)	High	8.1	CI/CD
CI-02	Untrusted fork-PR code built and executed on a large runner; PR-built artifacts uploaded	High	8.1	CI/CD
PB1-03	Destructive response actions (account disable/enable, quarantine, kill, delete, block, global prevent IOC) with no approval gate or allow-list	High	8.1	SOAR playbooks
PB2-01	OS command injection via ip_or_hostname in SSH investigate playbooks	High	8.0	SOAR playbooks
PB2-02	OS command injection via filepath in internal_host_ssh_log4j_respond	High	8.0	SOAR playbooks
PB2-04	Command injection and arbitrary file deletion via filePath in delete_detected_files (no approval gate)	High	8.0	SOAR playbooks
PB1-06	Hard-coded AWS IAM console password applied to every re-enabled user (public repository)	High	7.4	SOAR playbooks
PB2-10	Destructive automated response with no approval gate or protected-entity allow-list	High	7.4	SOAR playbooks
PB2-09	Attacker-steerable host shutdown and network isolation in the log4j respond pack	High	7.1	SOAR playbooks
DEP-01	Build toolchain dependency tree unlocked: 65 transitive packages float with no lockfile or hashes	Medium	6.8	Dependencies
DEP-05	All 28 GitHub Action references use mutable major tags; third-party actions run with PAT and AWS keys	Medium	6.7	Dependencies
CF-01	Unescaped input interpolated into SOAR REST _filter_* query strings (filter/parameter injection)	Medium	6.5	SOAR custom functions
CF-02	container_update mass-assignment via input_json	Medium	6.5	SOAR custom

ID	Title	Severity	CVSS	Area
	with no key allow-list			functions
CI-03	Splunkbase session token printed to CI logs by ta_update.py	Medium	6.5	CI/CD
PB2-08	Cleartext new password written into a SOAR container comment and action output	Medium	6.5	SOAR playbooks
PB2-06	SPL injection via untrusted CEF fields in Splunk run query templates (multiple playbooks)	Medium	6.3	SOAR playbooks
APP-01	SPL injection via weakly-quoted dashboard token $story$ (ESCU summary dashboard)	Medium	6.1	Splunk app and dashboards
APP-02	SPL injection via unquoted $policyname$/$eventcode$ tokens with auto-run on load (AppLocker dashboard)	Medium	6.1	Splunk app and dashboards
DEP-03	Splunk apps, attack data and container images downloaded without integrity verification from mutable locations	Medium	6.1	Dependencies
CF-03	TLS verification disabled on every custom-function REST call (30+ sites)	Medium	5.9	SOAR custom functions
PB2-07	Non-cryptographic PRNG used to generate the new Active Directory password	Medium	5.9	SOAR playbooks
DET-03	New-IP and spike detections write their own allow-list from attacker-influenced events (self-poisoning)	Medium	5.8	Detection content
PB2-05	Raw IP address passed as the whole Splunk/Carbon Black query (hunting.py)	Medium	5.8	SOAR playbooks
DEP-04	End-of-life components: Splunk 9.3 test image (EOL 2026-07-24), Python 3.10 CI runtime (EOL 2026-10-01), legacy SOAR exports, notebook stacks	Medium	5.6	Dependencies
CF-06	regex_filter_list uses prefix-anchored re.match for keep/drop filtering (allow-list bypass)	Medium	5.4	SOAR custom functions
CF-09	container_merge: unauthorised close/merge/evidence on attacker-influenced container IDs (IDOR)	Medium	5.4	SOAR custom functions
PB1-07	Attacker-controlled e-mail Message-ID drives tenant-wide search, delete and restore (MS Graph, Gmail)	Medium	5.4	SOAR playbooks
CF-04	zip_extract: uncapped decompression (zip-bomb DoS) and no temp cleanup; Zip-Slip not exploitable	Medium	5.3	SOAR custom functions
CF-05	ReDoS via user-supplied regex compiled in regex_filter_list and regex_split	Medium	5.3	SOAR custom functions
CF-07	regex_extract_email: polynomial backtracking and incorrect extraction of attacker-controlled domains	Medium	5.3	SOAR custom functions
CI-05	XML parsing of remote responses without defusedxml and HTTP requests without timeout (ta_update.py)	Medium	5.3	CI/CD
DET-02	SPL injection in potential_password_in_username via attacker-chosen username passed through map	Medium	5.3	Detection content
DET-04	Previously-seen baselines and their consumers form an allow-list poisoning path (30 files)	Medium	5.3	Detection content

ID	Title	Severity	CVSS	Area
DET-06	Case-sensitive match()/LIKE on attacker-controlled fields allows trivial bypass	Medium	5.3	Detection content
DET-07	Hard-coded placeholder domain internal_test_email.com ships as live detection logic (dead detections)	Medium	5.3	Detection content
PB1-08	Cisco Talos Untrusted verdict never maps to Malicious (unstrusted typo) - fail-open scoring	Medium	5.3	SOAR playbooks
PB1-05	ServiceNow sysparm_query (encoded query and URL parameter) injection via search_term	Medium	5.0	SOAR playbooks
DET-05	ES drilldown and risk-message $field$ tokens are substituted unquoted (1,971 detections; one uses |s)	Medium	4.3	Detection content
PB1-04	LDAP filter injection in AD LDAP Entity Attribute Lookup	Medium	4.2	SOAR playbooks
DEP-06	Licence compliance gaps: GPL-3.0-derived datasets redistributed under Apache-2.0 without notices; no NOTICE file; build tool licence unknown	Medium	—	Dependencies
CF-11	asset_get_attributes returns the full asset configuration (possible secret exposure)	Low	4.9	SOAR custom functions
PB2-15	Hostname interpolated into PowerShell file paths in internal_host_winrm_investigate	Low	4.4	SOAR playbooks
CF-08	regex_extract_ipv4: unanchored regex extracts wrong or partial IPs used for security decisions	Low	4.3	SOAR custom functions
CF-10	Object-reference inputs acted on without authorisation checks (artifact/indicator/evidence functions)	Low	4.3	SOAR custom functions
CI-08	External tools launched via partial executable path (PATH hijack) in pre-commit hook and scripts	Low	4.2	CI/CD
DET-16	DSDL/MLTK notebooks deserialise models with pickle and torch.load on stale libraries	Low	3.9	Detection content
DEP-10	Unpinned yamlfmt@latest, floating Go toolchain and sibling-directory executable probe in the pre-commit hook	Low	3.8	Dependencies
CF-14	url_parse: unhandled ValueError on malformed port	Low	3.7	SOAR custom functions
DEP-08	Splunk TA pins drift from Splunkbase and from each other (10 apps a major version behind)	Low	3.7	Dependencies
PB2-11	TLS certificate verification disabled on SOAR REST calls (verify=False)	Low	3.7	SOAR playbooks
SEC-03	Incomplete sanitisation of CloudTrail example logs leaks real AWS account IDs, access-key IDs, principals, roles and buckets	Low	3.7	Secrets
SEC-05	Personal data of employees and residential IP addresses embedded in sample logs	Low	3.7	Secrets
SEC-10	Internal infrastructure and cloud-tenant identifiers disclosed (hostnames, private URLs, tenant GUIDs, S3 buckets, playground endpoint)	Low	3.7	Secrets

ID	Title	Severity	CVSS	Area
CI-13	Missing top-level permissions: blocks and use of a personal access token where a scoped token would do	Low	3.5	CI/CD
PB1-10	Attacker-controlled values rendered unescaped into SOAR markdown notes	Low	3.5	SOAR playbooks
PB2-12	Allow-list bypass by self-tagging in aws_disable_user_accounts	Low	3.5	SOAR playbooks
CF-12	debug.py and passthrough.py log full input values (sensitive data to logs)	Low	3.3	SOAR custom functions
CI-09	Bare except: and broad exception handling swallow errors (validate_yaml.py, ta_update.py)	Low	3.3	CI/CD
APP-08	default.meta exports all knowledge objects system-wide with admin/sc_admin write (tampering surface)	Low	3.1	Splunk app and dashboards
CF-13	Validation via assert statements (stripped under python -O)	Low	3.1	SOAR custom functions
CF-15	Missing None/type guards and empty-cell substring logic bug (custom_list_value_in_strings)	Low	3.1	SOAR custom functions
CF-16	base64_decode: fail-open decoding, NUL stripping and non-ASCII drop	Low	3.1	SOAR custom functions
CF-18	zip_extract: unvalidated vault_id used to build the temp extraction path	Low	3.1	SOAR custom functions
CI-10	PR-controlled detection name/exception rendered unescaped into the GitHub step summary	Low	3.1	CI/CD
DEP-09	SOAR playbooks declare no app versions and depend on unversioned community functions and platform-bundled libraries	Low	3.1	Dependencies
DEP-12	Dependency-update automation does not cover pip or docker ecosystems; no SBOM produced with releases	Low	3.1	Dependencies
DET-08	Malformed drilldown SPL in Copilot and Ollama detections (unbalanced quotes)	Low	3.1	Detection content
PB1-09	CrowdStrike Dynamic Analysis URL scoring is dead code or crash-prone (wrong datapaths, zip order)	Low	3.1	SOAR playbooks
PB1-11	Destructive leaf playbooks read raw unfiltered inputs, defeating the preceding filter	Low	3.1	SOAR playbooks
PB1-13	Account dispatch report/tagging logic flaws (only first user tagged, duplicated rows, None crash)	Low	3.1	SOAR playbooks
PB2-13	Hard-coded placeholder email recipient and sender in email_notification_for_malware	Low	3.1	SOAR playbooks
PB2-14	Unsanitised artifact data in notification email body	Low	3.1	SOAR playbooks
DET-12	map subsearches with non-attacker-controlled $token$ substitution (defence-in-depth)	Low	2.6	Detection content
PB1-12	Unguarded dict/list/regex access on connector and artifact data (crash / DoS)	Low	2.6	SOAR playbooks

ID	Title	Severity	CVSS	Area
PB1-14	Shallow-copy bug mislabels interaction artifacts in Automated Email Investigation	Low	2.6	SOAR playbooks
PB1-15	Super-linear regex on endpoint-controlled RTR stdout and invalid escape sequences	Low	2.6	SOAR playbooks
CI-15	Unbounded $ref recursion in validate_response_templates.py	Low	2.3	CI/CD
DET-09	Drilldown tokens not produced by the detection search (102 dead pivots)	Low	2.3	Detection content
APP-07	Lookup CSV cells beginning with formula characters (export-boundary CSV injection surface only)	Low	2.0	Splunk app and dashboards
APP-03	Logic defect: missing leading pipe in remove_valid_domains subsearch	Low	—	Splunk app and dashboards
APP-04	Over-broad placeholder filter macros (search *) and permissive shipped allow-lists	Low	—	Splunk app and dashboards
APP-05	Lookup WILDCARD match_type references a column absent from the CSV (suspicious_ports_list)	Low	—	Splunk app and dashboards
APP-06	Stale and unattributed threat-intelligence lookups (SSLBL, dynamic DNS providers)	Low	—	Splunk app and dashboards
APP-09	Example identity lookups shipped as defaults (domain_controllers, domain_admins)	Low	—	Splunk app and dashboards
APP-12	Response templates: unquoted src_user SPL token and instance-specific asset IDs	Low	—	Splunk app and dashboards
DEP-07	contentctl-ng has no licence metadata, placeholder description and a non-public source repository	Low	—	Dependencies
SEC-08	Secret-prevention controls are weak: no scanner in CI, inert pre-commit hook, incomplete .gitignore	Low	—	Secrets
PB1-16	phantom.debug logs full observable and result structures (investigation data to logs)	Informational	1.8	SOAR playbooks
APP-10	Lookup CSV header hygiene: UTF-8 BOM and leading-space field names	Informational	—	Splunk app and dashboards
APP-13	JSON schemas leave SPL/playbook/cron fields unconstrained; RemovedContent schema lacks additionalProperties:false	Informational	—	Splunk app and dashboards
APP-14	Documentation instructs bypassing pre-commit hooks (git commit --no-verify); yamllint scoped to detections only	Informational	—	Splunk app and dashboards
CF-17	Deprecated naive-UTC datetime APIs and epoch truncation	Informational	—	SOAR custom functions

ID	Title	Severity	CVSS	Area
CI-11	.github/workflows/labeler.yml uses pull_request_target safely (positive control, informational)	Informational	—	CI/CD
CI-12	OpenAPI example values flagged by secret scanners are placeholders (informational)	Informational	—	CI/CD
CI-14	Hard-coded admin password for the throw-away CI Splunk container (informational)	Informational	—	CI/CD
CI-16	bump-yaml.sh follows secure shell practices (positive note)	Informational	—	CI/CD
DET-13	Cleartext http:// references and malformed references in shipped content	Informational	—	Detection content
DET-14	66 production detections validated only by disabled manual tests (attack_data provenance otherwise clean)	Informational	—	Detection content
DET-15	Deprecated and removed content remains in tree; contentctl excludes it from live searches (positive control)	Informational	—	Detection content
DET-18	Playbook metadata references non-existent or mismatched JSON filenames	Informational	—	Detection content
DET-19	Detection identifiers unique; 127 detections with empty data_source (schema conformance otherwise clean)	Informational	—	Detection content
PB1-17	Correctness and dead-code defects (misrouted enrichment dispatch, wrong keys, column mismatch)	Informational	—	SOAR playbooks
PB1-18	Analyst-prompt design weaknesses in the Search-and-Purge playbooks	Informational	—	SOAR playbooks
PB2-16	Bare except and unguarded dictionary/division in result normalisation blocks	Informational	—	SOAR playbooks
PB2-17	Legacy python_version and Classic-editor playbooks carry the ungated destructive patterns	Informational	—	SOAR playbooks
SEC-04	Expired AWS STS temporary credential committed in a sample log	Informational	—	Secrets
SEC-09	Third-party reputation-service URL with alliance key parameters in SOAR playbook sample values	Informational	—	Secrets

NIST SP 800-53 Revision 5

Findings are mapped to the NIST SP 800-53 Rev. 5 controls that address them.

Control	Findings	Count
AC-3 Access Enforcement	APP-08, CF-02, CF-09, CF-10, DET-04, PB1-01, PB1-03, PB1-18, PB2-09, PB2-10, PB2-12	11
AC-6 Least Privilege	APP-08, CF-11, CI-13, DEP-02, DEP-05, PB1-01, PB1-03, PB1-05, PB2-01, PB2-02, PB2-04, PB2-06, PB2-12, SEC-03	14

Control	Findings	Count
AU-6 Audit Record Review, Analysis, and Reporting	DET-03, DET-04	2
AU-9 Protection of Audit Information	CF-12, CI-03, PB1-16, PB2-08	4
AU-12 Audit Record Generation	DET-03, DET-04	2
CM-2 Baseline Configuration	CI-02, DET-15, DET-18	3
CM-3 Configuration Change Control	APP-14, CI-01, SEC-08	3
CM-5 Access Restrictions for Change	APP-08, PB1-02, PB1-03, PB2-09, PB2-10	5
CM-6 Configuration Settings	APP-04, APP-05, APP-09, APP-12, DET-07, PB2-13	6
CM-7 Least Functionality	CI-08, CI-13, DEP-05, DEP-10	4
CM-8 System Component Inventory	DEP-01, DEP-02, DEP-03, DEP-04, DEP-06, DEP-07, DEP-08, DEP-09, DEP-12	9
IA-5 Authenticator Management	CI-03, PB1-06, PB2-07, PB2-08, SEC-04	5
IA-5(7) Authenticator Management | No Embedded Unencrypted Static Authenticators	DEP-02, PB1-06	2
IR-4 Incident Handling	PB1-03, PB1-07, PB1-18	3
RA-5 Vulnerability Monitoring and Scanning	APP-06, DEP-01, DEP-12, SEC-03	4
SA-11 Developer Testing and Evaluation	APP-03, APP-05, APP-10, APP-13, APP-14, DET-14, SEC-03, SEC-05, SEC-08, SEC-09, SEC-10	11
SA-15 Development Process, Standards, and Tools	APP-14, CI-01, CI-02, SEC-08	4
SA-22 Unsupported System Components	DEP-01, DEP-02, DEP-03, DEP-04, DEP-05, DEP-06, DEP-08, DEP-09, DEP-10, PB2-17	10
SC-5 Denial-of-Service Protection	CF-04, CF-05, CF-07, CI-05, PB1-15	5
SC-7 Boundary Protection	CI-02	1
SC-8 Transmission Confidentiality and Integrity	CF-03, DET-13, PB2-11	3
SC-13 Cryptographic Protection	PB2-07	1
SC-18 Mobile Code	APP-07	1
SC-28 Protection of Information at Rest	CF-11, CI-03, SEC-03, SEC-04, SEC-05, SEC-10	6
SI-2 Flaw Remediation	CF-17, DEP-01, DEP-02, DEP-04, DEP-08, DEP-12	6
SI-4 System Monitoring	APP-04, DET-02, DET-03, DET-04, DET-06, DET-07, DET-08, PB1-08, PB2-10	9
SI-7 Software, Firmware, and Information Integrity	APP-08, CI-08, DEP-03, DET-16, SEC-08, SEC-10	6
SI-10 Information Input Validation	APP-01, APP-02, APP-07, APP-10, APP-12, APP-13, CF-01, CF-02, CF-04, CF-05, CF-06, CF-07, CF-08, CF-14, CF-15, CF-16, CF-18, CI-01, CI-05, CI-10, CI-15, DET-02, DET-05, DET-12, PB1-01, PB1-02, PB1-04, PB1-05, PB1-07, PB1-08, PB1-09, PB1-10, PB1-11, PB1-15, PB2-01, PB2-02, PB2-03, PB2-04, PB2-05, PB2-06, PB2-14, PB2-	42

Control	Findings	Count
	15	
SI-11 Error Handling	CF-12, CF-13, CF-14, CI-09, PB1-12, PB1-13, PB1-14, PB1-16, PB2-16	9
SR-3 Supply Chain Controls and Processes	DEP-01, DEP-02, DEP-05, DEP-06, DEP-07, DEP-09	6
SR-4 Provenance	DEP-01, DEP-02, DEP-03, DEP-05, DEP-06, DEP-07, DEP-10, DEP-12, SEC-10	9
SR-11 Component Authenticity	DEP-03, DET-16	2
AC-4	APP-01, APP-02	2
PT-2	SEC-05	1
SA-9	DEP-07	1
SI-5	APP-06	1

NIST Cybersecurity Framework 2.0

Findings are mapped to CSF 2.0 function and category.

Category	Findings	Count
GV.SC Govern - Cybersecurity Supply Chain Risk Management	APP-14, DEP-01, DEP-02, DEP-03, DEP-05, DEP-06, DEP-07, DEP-09, DEP-12, SEC-10	10
ID.AM Identify - Asset Management	DEP-04, DEP-06, DEP-08, DEP-09, DEP-12, DET-19	6
ID.RA Identify - Risk Assessment	APP-06, DEP-01, SEC-03	3
ID.IM Identify - Improvement	DET-14	1
PR.AA Protect - Identity Management, Authentication, and Access Control	APP-08, CF-09, CF-10, CI-03, CI-13, DEP-02, PB1-03, PB1-06, PB1-18, PB2-07, PB2-09, PB2-10, PB2-12	13
PR.DS Protect - Data Security	CF-03, CF-11, CF-12, CI-03, DEP-03, DET-02, PB2-08, PB2-11, SEC-03, SEC-04, SEC-05, SEC-10	12
PR.PS Protect - Platform Security	APP-01, APP-02, APP-03, APP-04, APP-05, APP-06, APP-07, APP-09, APP-10, APP-12, APP-13, APP-14, CF-01, CF-02, CF-04, CF-05, CF-06, CF-07, CF-08, CF-13, CF-14, CF-15, CF-18, CI-01, CI-02, CI-05, CI-08, CI-09, CI-10, CI-15, DEP-01, DEP-02, DEP-04, DEP-05, DEP-08, DEP-10, DET-05, DET-12, DET-16, PB1-01, PB1-02, PB1-03, PB1-04, PB1-05, PB1-07, PB1-10, PB1-11, PB1-12, PB1-13, PB1-14, PB1-15, PB1-16, PB2-01, PB2-02, PB2-03, PB2-04, PB2-05, PB2-06, PB2-13, PB2-14, PB2-15, PB2-16, PB2-17, SEC-08	64
DE.CM Detect - Continuous Monitoring	APP-04, APP-08, CI-01, DET-02, DET-03, DET-	11

Category	Findings	Count
	04, DET-06, DET-07, DET-08, DET-09, SEC-08	
DE.AE Detect - Adverse Event Analysis	CF-16, DET-03, DET-04, PB1-08, PB1-09, PB1-16	6
RS.MI Respond - Incident Mitigation	PB1-03, PB2-09, PB2-10	3